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United States Patent	12398342
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Lee; Su Jin et al.

Process liquid composition for lithography and pattern forming method using same

Abstract

Proposed is a process liquid composition for improving a lifting defect level of a photoresist pattern containing a surfactant and for reducing the number of defects of the photoresist pattern, the composition containing a surfactant and having a surface tension of 40 mN/m or less and a contact angle of 60° or smaller in the photoresist pattern having hydrophobicity represented by a contact angle of 70° or greater of water with respect to a photoresist surface in a photoresist pattern process.

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Appl. No.: **17/626357**

Filed (or PCT Filed): **June 24, 2020**

PCT No.: **PCT/KR2020/008141**

PCT Pub. No.: **WO2021/010609**

PCT Pub. Date: January 21, 2021

Prior Publication Data

Document Identifier	Publication Date
US 20220251472 A1	Aug. 11, 2022

Foreign Application Priority Data

KR	10-2019-0086818	Jul. 18, 2019

Publication Classification

Int. Cl.: **C11D1/00** (20060101); **C11D1/04** (20060101); **C11D1/12** (20060101); **C11D1/14** (20060101); **C11D1/34** (20060101); **C11D3/24** (20060101); **C11D3/30** (20060101); **G03F7/00** (20060101); **G03F7/20** (20060101); **G03F7/32** (20060101)

U.S. Cl.:

CPC **C11D1/004** (20130101); **C11D1/04** (20130101); **C11D1/12** (20130101); **C11D1/146** (20130101); **C11D1/345** (20130101); **C11D3/245** (20130101); **C11D3/30** (20130101); **G03F7/2004** (20130101); **G03F7/322** (20130101); **G03F7/70925** (20130101); C11D2111/22 (20240101)

Field of Classification Search

CPC: C11D (1/146); C11D (1/12); C11D (1/345); C11D (3/245); C11D (3/30)

USPC: 510/175; 510/176; 510/504

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Background/Summary

TECHNICAL FIELD

(1) The present invention relates to a process liquid composition for alleviating a lifting defect level of a photoresist pattern and for reducing the number of defects of the photoresist pattern, the photoresist pattern having hydrophobicity represented by a contact angle of 70° or greater of water with respect to a photoresist surface in a photoresist pattern process, and to a method of forming a photoresist pattern using the process liquid composition.

BACKGROUND ART

(2) Generally, a semiconductor is manufactured by a lithographic process in which exposure light is infrared light in a wavelength of 193 nm, 248 nm, 365 nm, or the like. There is intense competition among semiconductor manufacturers for reduction in a critical dimension (hereinafter referred to as a CD).

(3) Accordingly, the finer pattern is to be formed, the narrower wavelength a light source needs to produce. At the present time, a lithographic technology using an extreme ultraviolet (EUV in a wavelength of 13.5 nm) is actively employed. A narrower wavelength may be realized using this lithographic technology.

(4) However, the resistance of EUV photoresist to etching is not yet improved, and thus a photoresist pattern having a high aspect ratio still needs to be used. Accordingly, a pattern lifting defect occurs easily during development, and the number of defects is increased. Consequently, a process margin is greatly reduced in a manufacturing process.

(5) To solve this problem, there is a demand to develop the technology for alleviating a level of a lifting defect that occurs while forming a fine pattern and for reducing the number of defects. The best way to alleviate a pattern lifting defect level and reduce the number of defects may be to improve photoresist performance. However, there is a need to consider a situation where, in practice, it is difficult to develop new photoresist having performance that is satisfactory in terms of all aspects.

(6) There is still a need to develop new photoresist. However, attempts have been made to alleviate the pattern lifting defect level and reduce the number of defects in ways other than addressing this need.

DISCLOSURE

Technical Problem

(7) An objective of the present invention is to develop a process liquid composition for alleviating a level of a pattern lifting defect and reducing the number of defects, the pattern lifting defect

occurring after developing photoresist having hydrophobicity represented by a contact angle of 70° or greater of water with respect to a photoresist surface, and to develop a method of forming a photoresist pattern using the process liquid composition.

Technical Solution

(8) Various surfactants are used to manufacture a water-based process liquid composition that is used during a developing process. However, according to the present invention, an effective process liquid composition was manufactured using a fluorine-based surfactant and a hydrocarbon-based anionic surfactant.

(9) The use of a hydrocarbon-based non-ionic surfactant with a property like hydrophobicity in manufacturing the water-based process liquid composition in which ultra-pure water is mostly contained may lead to forming a hydrophobic sidewall of a photoresist and thus reducing pattern melting or collapse. However, in this case, the hydrocarbon-based non-ionic surfactants have a strong tendency to agglomerate, resulting in preventing a property of the process liquid composition from being uniform. Theretofore, there is a likelihood that the agglomerating hydrocarbon-based non-ionic surfactants will cause a defect while the process liquid composition is in use. That is, the use of the hydrocarbon-based non-ionic surfactant requires an increase in the usage amount thereof for reducing the pattern melting. Thus, there is a concern that photoresist will be damaged. In addition, the excessive use of an unsuitable surfactant for the purpose of reducing surface tension of the process liquid composition to reduce a capillary force may lead to the pattern melting and rather may further cause the pattern collapse.

(10) In addition, in the case of a hydrocarbon-based cationic surfactant, an active group dissociates into a cation in an aqueous solution, and it is rarely ensured that metal is formed. Thus, there is a concern that a serious defect will be caused to occur in a lithographic process.

(11) According to the present invention, it was verified that the use of the fluorine-based surfactant and the hydrocarbon-based anionic surfactant achieved the noticeable effect of alleviating the pattern lifting defect level and reducing the number of defects. The surface tension and the contact angle, which were much more decreased than in the hydrocarbon-based non-ionic surfactant, increased penetrability and spreadability, leading to contribution to formation of a fine pattern. It was recognized that this contribution resulted in the noticeable effect.

(12) Tetramethylammonium hydroxide is diluted with pure water to a predetermined concentration (2.38% by weight of tetramethylammonium hydroxide is mixed with 97.62% by weight of water for use in most of the photolithographic developing processes) for use as a representative developing solution that is currently used in most of the photolithographic developing processes.

(13) It was verified that a pattern lifting defect was caused in a case where, in a photolithographic process, a photoresist pattern having hydrophobicity represented by a contact angle of 70° or greater of water with respect to a photoresist surface was successively cleaned only with pure water after being developed. Furthermore, it was verified that, in the photolithographic process, the pattern collapse was also caused in a case where a process liquid composition resulting from tetramethylammonium hydroxide being contained in pure water was successively applied after developing or in a case where pure water was successively applied.

(14) It could be estimated that the pattern collapse was caused because the process liquid composition containing tetramethylammonium hydroxide weakened the exposed fine pattern and because the capillary force was great and was non-uniform.

(15) Therefore, in order to prevent the exposed-pattern collapse and to reduce the line width roughness (LWR) and the number of defects additionally required in a process, there is a need to conduct a study on an alkali substance that exerts a relatively weaker force on the exposed pattern than tetramethylammonium hydroxide.

(16) According to the present invention, it was verified that, in a case where tetraethylammonium hydroxide, tetrapropylammonium hydroxide, and tetrabutylammonium hydroxide was used among alkali substances, not only was the pattern collapse prevented and the LWR, but the number of

defects was also reduced.
(17) According to a desirable first embodiment of the present invention, there is provided a process liquid composition for alleviating a level of a lifting defect of a photoresist pattern and for reducing the number of lifting defects of the photoresist pattern, the composition containing a surfactant and having a surface tension of 40 millinewton/meter (mN/m= 1/1000 newton/meter) or less and a contact angle of 60° or smaller in the photoresist pattern having hydrophobicity represented by a contact angle of 70° or greater of water with respect to a photoresist surface in a photoresist pattern process.
(18) According to a more desirable second embodiment of the present invention, there is provided a process liquid composition for alleviating a level of a lifting defect of a photoresist pattern and for reducing the number of lifting defects of the photoresist pattern, the lifting defect occurring during photoresist developing, the process liquid composition containing: 0.00001 to 0.1% by weight of a fluorine-based surfactant; 0.0001 to 0.1% by weight of a hydrocarbon-based anionic surfactant; 0.0001 to 0.1% by weight of an alkali substance; and 99.7 to 99.99979% by weight of water.
(19) According to the most desirable third embodiment of the prevent invention, there is a process liquid composition for alleviating a level of a lifting defect of a photoresist pattern and for reducing the number of lifting defects of the photoresist pattern, the lifting defect occurring during photoresist developing, the process liquid composition containing: 0.00001 to 0.1% by weight of a fluorine-based surfactant; 0.001 to 0.1% by weight of a hydrocarbon-based anionic surfactant; 0.001 to 0.1% by weight of an alkali substance; and 99.7 to 99.9979% by weight of water, the composition having a surface tension of 40 mN/m or less and a contact angle of 60° or smaller.
(20) In the process liquid composition according to any one of the first to third embodiments, the fluorine-based surfactant may be selected from the group consisting of fluoroacryl carboxylate, fluoroalkyl ether, fluoroalkylene ether, fluoroalkyl sulfate, fluoroalkyl phosphate, fluoroacryl co-polymer, fluoro co-polymer, perfluorinated acid, perfluorinated carboxylate, perfluorianted sulfonate, and mixtures thereof.
(21) In the process liquid composition according to any one of the second to third embodiments, wherein the hydrocarbon-based anionic surfactant may be selected from the group consisting of ammonium salt of polycarboxylic acid, sulfonate salt, sulfate ester salt, phosphoric acid ester salt, and mixtures thereof.
(22) In the process liquid composition according to any one of the first to third embodiments, wherein the alkali substance may be selected from the group consisting of tetraethylammonium hydroxide, tetrapropylammonium hydroxide, tetrabutylammonium hydroxide, and mixtures thereof.
(23) According to an aspect of the present invention, there is provided a method of forming a photoresist pattern, the method including: (a) a step of dispensing photoresist on a semiconductor substrate and forming a photoresist film; (b) a step of exposing the photoresist film to light, developing the photoresist film, and forming a photoresist pattern; and (c) a step of cleaning the photoresist pattern with the process liquid composition.
(24) It was thought that the pattern collapse was caused by the capillary force occurring between patterns when the patterns were cleaned with pure water after developing. However, it was experimentally recognized that only the reduction of the capillary force could neither completely prevent the pattern collapse nor reduce the number of the lifting defects.
(25) The excessive use of the unsuitable surfactant for the purpose of reducing the surface tension of the process liquid composition to reduce the capillary force may lead to the pattern melting and rather may further cause the pattern collapse or increase the number of lifting defects.
(26) In order to alleviate the level of the pattern lifting defect and reduce the number of the pattern lifting defects, it is important to select a surfactant that reduces the surface tension of the process liquid composition and at the same time prevents the melting of the photoresist pattern.
(27) The process liquid composition according to the present invention exerts an enhancing effect

on the photoresist and particularly achieves the effect of alleviating the level of the pattern lifting defect and the number of the pattern lifting defects, the pattern lifting defect occurring while developing photoresist having hydrophobicity represented by a contact angle of 70° or greater of water with respect to a photoresist surface.

Advantageous Effects

(28) The process liquid composition according to the present invention achieves the effect of alleviating the level of the pattern lifting defect and the number of the pattern lifting defects, the effect that cannot be achieved only with photoresist when a pattern is formed using the photoresist having hydrophobicity represented by a contact angle of 70° or greater of water with respect to a photoresist surface. The photoresist forming method including the step of cleaning the photoresist pattern with the process liquid composition can achieve the effect of greatly reducing manufacturing cost.

Description

BEST MODE

(1) The present invention will be described in more detail below.

(2) The present invention, which is the result of conducting much research over a long period of time, relates to a "process liquid composition for alleviating a lifting defect level of a photoresist pattern and reducing the number of defects of the photoresist, the process liquid composition containing: 0.00001 to 0.1% by weight of a fluorine-based surfactant selected from the group consisting of fluoroacryl carboxylate, fluoroalkyl ether, fluoroalkylene ether, fluoroalkyl sulfate, fluoroalkyl phosphate, fluoroacryl co-polymer), fluoro co-polymer, perfluorinated acid, perfluorinated carboxylate, perfluorianted sulfonate, and mixtures thereof; 0.0001 to 0.1% by weight of an anionic surfactant selected from the group consisting of ammonium salt of polycarboxylic acid, sulfonate salt, sulfate ester salt, phosphoric acid ester salt, and mixtures thereof; 0.0001 to 0.1% by weight of an alkali substance selected from the group consisting of tetraethylammonium hydroxide, tetrapropylammonium hydroxide, tetrabutylammonium hydroxide, and mixtures thereof; and 99.7 to 99.99979% by weight of water". Composition components of the process liquid composition according to the present invention and a composition ratio between the components thereof were specified as shown in Embodiments 1 to 60. Composition components and a composition ratio that were in contrast with the above-mentioned composition components and composition ratio, respectively, were specified as shown in Comparative Examples 1 to 12.

(3) Desired embodiments of the present invention and comparative examples for comparison therewith will be described below. However, the desired embodiments described below of the present invention are only exemplary, and the present invention is not limited thereto.

MODE FOR INVENTION

Embodiment 1

(4) A process liquid composition for alleviating a collapse level of a photoresist pattern, which contains 0.001% by weight of fluoroacryl carboxylate, 0.01% by weight of ammonium salt of polycarboxylic acid, and 0.005% by weight of tetrabutylammonium hydroxide, was manufactured using the following method.

(5) 0.001% by weight of fluoroacryl carboxylate, 0.01% by weight of ammonium salt of polycarboxylic acid, and 0.005% by weight of tetrabutylammonium hydroxide were added into a remaining amount of distilled water and stirred for 5 hours. Then, the resulting liquid was caused to pass through a filter with a size of 0.01 μm to remove fine-sized soluble-solid impurities. In this manner, the process liquid composition for alleviating the collapse level of the photoresist pattern was manufactured.

Embodiments 2 to 60

(6) Process liquid compositions for alleviating a defect level of a photoresist pattern that was the same as a defect level of a photoresist pattern in Embodiment 1 were manufactured according to composition components and component ratios therebetween that were specified as shown in Tables 1 to 12.

Comparative Example 1

(7) Usually, distilled water that was to be used as a cleaning solution in the last process among semiconductor manufacturing processes was prepared.

Comparative Examples 2 to 12

(8) For comparison with embodiments, process liquid compositions were manufactured, as in Embodiment 1, according to the composition components and the component ratios therebetween that were specified as shown in Tables 1 to 12.

Experimental Examples 1 to 60 and Comparative Experimental Examples 1 to 12

(9) Measurements of pattern lifting defect levels and number-of-defects reduction ratios were performed on silicon wafers, patterns on which were formed in Embodiments 1 to 60 and Comparative Examples 1 to 12. The measurements are described as Experimental Examples 1 to 60 and Comparative Experimental Examples 1 to 12. The results of the measurements are shown in Table 13.

(10) (1) Verification of Pattern Lifting Prevention

(11) After exposure energy and focus were split, among a total of 89 blocks, the number of blocks in which a pattern did not collapse was measured using a critical dimension-scanning electron microscope (CD-SEM, manufactured by Hitachi, Ltd).

(12) (2) Number-of-Lifting-Defects Reduction Ratio

(13) Counting of the number A of defects was performed on a photoresist pattern that was rinsed with each process liquid composition sample, using surface defect observation equipment (manufactured by KLA-Tencor Corporation). A value of 100 was assigned to the number B of defects that resulted when the photoresist pattern was rinsed only with pure water. Then, the number A of defects was expressed as a ratio to the number B of defects, that is, as (A/B)×100.

(14) The number of defects that resulted when rinsing was performed only with pure water was defined as 100. The degree to which the number of defects was decreased (improved) or increased (degraded) when compared with the number of defects resulting from rinsing only with pure water was expressed as a reduction ratio.

(15) (3) Transparency

(16) Transparency of the manufactured process liquid composition was checked with the naked eye and was marked as a transparent or opaque process liquid composition.

(17) (4) Surface Tension and Contact Angle

(18) A surface tension and a contact angle of each of the process liquid compositions were measured using a surface tension measuring instrument [the K100 Force Tensiometer manufactured by KRÜSS GmbH] and a contact angle measuring instrument [the DSA-100 Drop Shape Analyzer manufactured by KRÜSS GmbH].

(19) TABLE-US-00001 TABLE 1 Surfactant Surfactant Alkali substance Distilled water Content Content Content Content (% by (% by (% by (% by Name weight) Name weight) Name weight) Name weight) Embodiment 1 Fluoroacryl 0.001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 carboxylate polycarboxylic acid hydroxide water Embodiment 2 Fluoroalkyl 0.001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 ether polycarboxylic acid hydroxide water Embodiment 3 Fluoroalkylene 0.001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 ether polycarboxylic acid hydroxide water Embodiment 4 Fluoroalkyl 0.001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 sulfate polycarboxylic acid hydroxide water Embodiment 5 Fluoroalkyl 0.001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 phosphate polycarboxylic acid hydroxide water Embodiment 6 Fluoroacryl 0.001 Ammonium salt of 0.01 Tetrabutylammonium 0.005

Distilled 99.9840 co-polymer polycarboxylic acid hydroxide water Embodiment 7 Fluoro 0.001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 co-polymer polycarboxylic acid hydroxide water Embodiment 8 Perfluorinated 0.001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 acid polycarboxylic acid hydroxide water Embodiment 9 Perfluorinated 0.001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 carboxylate polycarboxylic acid hydroxide water Embodiment 10 Perfluorianted 0.001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 sulfonate polycarboxylic acid hydroxide water Comparative — — — — — — Distilled 100 Example 1 water Comparative — — — — Tetrabutylammonium 0.005 Distilled 99.9950 Example2 hydroxide water

(20) TABLE-US-00002 TABLE 2 Surfactant Surfactant Alkali substance Distilled water Content Content Content Content (% by (% by (% by (% by Name weight) Name weight) Name weight) Name weight) Embodiment 11 Fluoroacryl 0.001 Sulfonate 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 carboxylate salt hydroxide water Embodiment 12 Fluoroalkyl 0.001 Sulfonate 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 ether salt hydroxide water Embodiment 13 Fluoroalkylene 0.001 Sulfonate 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 ether salt hydroxide water Embodiment 14 Fluoroalkyl 0.001 Sulfonate 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 sulfate salt hydroxide water Embodiment 15 Fluoroalkyl 0.001 Sulfonate 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 phosphate salt hydroxide water Embodiment 16 Fluoroacryl 0.001 Sulfonate 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 co-polymer salt hydroxide water Embodiment 17 Fluoro 0.001 Sulfonate 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 co-polymer salt hydroxide water Embodiment 18 Perfluorinated 0.001 Sulfonate 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 acid salt hydroxide water Embodiment 19 Perfluorinated 0.001 Sulfonate 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 carboxylate salt hydroxide water Embodiment 20 Perfluorianted 0.001 Sulfonate 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 sulfonate salt hydroxide water

(21) TABLE-US-00003 TABLE 3 Surfactant Surfactant Alkali substance Distilled water Content Content Content Content (% by (% by (% by (% by Name weight) Name weight) Name weight) Name weight) Embodiment 21 Fluoroacryl 0.00001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.98499 carboxylate polycarboxylic acid hydroxide water Embodiment 22 Fluoroacryl 0.0001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9849 carboxylate polycarboxylic acid hydroxide water Embodiment 1 Fluoroacryl 0.001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 carboxylate polycarboxylic acid hydroxide water Embodiment 23 Fluoroacryl 0.01 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9750 carboxylate polycarboxylic acid hydroxide water Embodiment 24 Fluoroacryl 0.1 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.8850 carboxylate polycarboxylic acid hydroxide water Comparative Fluoroacryl 1 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 98.9850 Example 3 carboxylate polycarboxylic acid hydroxide water

(22) TABLE-US-00004 TABLE 4 Surfactant Surfactant Alkali substance Distilled water Content Content Content Content (% by (% by (% by (% by Name weight) Name weight) Name weight) Name weight) Embodiment 25 Fluoroalkyl 0.00001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.98499 ether polycarboxylic acid hydroxide water Embodiment 26 Fluoroalkyl 0.0001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9849 ether polycarboxylic acid hydroxide water Embodiment 2 Fluoroalkyl 0.001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 ether polycarboxylic acid hydroxide water Embodiment 27 Fluoroalkyl 0.01 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9750 ether polycarboxylic acid hydroxide water Embodiment 28 Fluoroalkyl 0.1 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.8850 ether polycarboxylic acid hydroxide water Comparative Fluoroalkyl 1 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 98.9850 Example 4 ether polycarboxylic acid hydroxide water

(23) TABLE-US-00005 TABLE 5 Surfactant Surfactant Alkali substance Distilled water Content

Content Content Content (% by (% by (% by (% by Name weight) Name weight) Name weight) Name weight) Embodiment 29 Fluoroalkylene 0.00001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.98499 ether polycarboxylic acid hydroxide water Embodiment 30 Fluoroalkylene 0.0001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9849 ether polycarboxylic acid hydroxide water Embodiment 3 Fluoroalkylene 0.001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 ether polycarboxylic acid hydroxide water Embodiment 31 Fluoroalkylene 0.01 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9750 ether polycarboxylic acid hydroxide water Embodiment 32 Fluoroalkylene 0.1 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.8850 ether polycarboxylic acid hydroxide water Comparative Fluoroalkylene 1 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 98.9850 Example 5 ether polycarboxylic acid hydroxide water

(24) TABLE-US-00006 TABLE 6 Surfactant Surfactant Alkali substance Distilled water Content Content Content Content (% by (% by (% by (% by Name weight) Name weight) Name weight) Name weight) Embodimen 33 Fluoroalkyl 0.00001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.98499 sulfate polycarboxylic acid hydroxide water Embodiment 34 Fluoroalkyl 0.0001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9849 sulfate polycarboxylic acid hydroxide water Embodiment 4 Fluoroalkyl 0.001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 sulfate polycarboxylic acid hydroxide water Embodiment 35 Fluoroalkyl 0.01 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9750 sulfate polycarboxylic acid hydroxide water Embodiment 36 Fluoroalkyl 0.1 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.8850 sulfate polycarboxylic acid hydroxide water Comparative Fluoroalkyl 1 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 98.9850 Example 6 sulfate polycarboxylic acid hydroxide water

(25) TABLE-US-00007 TABLE 7 Surfactant Surfactant Alkali substance Distilled water Content Content Content Content (% by (% by (% by (% by Name weight) Name weight) Name weight) Name weight) Embodiment 37 Fluoroalkyl 0.00001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.98499 phosphate polycarboxylic acid hydroxide water Embodiment 38 Fluoroalkyl 0.0001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9849 phosphate polycarboxylic acid hydroxide water Embodiment 5 Fluoroalkyl 0.001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 phosphate polycarboxylic acid hydroxide water Embodiment 39 Fluoroalkyl 0.01 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9750 phosphate polycarboxylic acid hydroxide water Embodiment 40 Fluoroalkyl 0.1 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.8850 phosphate polycarboxylic acid hydroxide water Comparative Fluoroalkyl 1 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 98.9850 Example 7 phosphate polycarboxylic acid hydroxide water

(26) TABLE-US-00008 TABLE 8 Surfactant Surfactant Alkali substance Distilled water Content Content Content Content (% by (% by (% by (% by Name weight) Name weight) Name weight) Name weight) Embodiment 41 Fluoroacryl 0.00001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.98499 co-polymer polycarboxylic acid hydroxide water Embodiment 42 Fluoroacryl 0.0001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9849 co-polymer polycarboxylic acid hydroxide water Embodiment 6 Fluoroacryl 0.001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 co-polymer polycarboxylic acid hydroxide water Embodiment 43 Fluoroacryl 0.01 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9750 co-polymer polycarboxylic acid hydroxide water Embodiment 44 Fluoroacryl 0.1 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.8850 co-polymer polycarboxylic acid hydroxide water Comparative Fluoroacryl 1 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 98.9850 Example 8 co-polymer polycarboxylic acid hydroxide water

(27) TABLE-US-00009 TABLE 9 Surfactant Surfactant Alkali substance Distilled water Content Content Content Content (% by (% by (% by (% by Name weight) Name weight) Name weight)

Name weight) Embodiment 45 Fluoro 0.00001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.98499 co-polymer polycarboxylic acid hydroxide water Embodiment 46 Fluoro 0.0001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9849 co-polymer polycarboxylic acid hydroxide water Embodiment 7 Fluoro 0.001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 co-polymer polycarboxylic acid hydroxide water Embodiment 47 Fluoro 0.01 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9750 co-polymer polycarboxylic acid hydroxide water Embodiment 48 Fluoro 0.1 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.8850 co-polymer polycarboxylic acid hydroxide water Comparative Fluoro 1 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 98.9850 Example 9 co-polymer polycarboxylic acid hydroxide water

(28) TABLE-US-00010 TABLE 10 Surfactant Surfactant Alkali substance Distilled water Content Content Content Content (% by (% by (% by (% by Name weight) Name weight) Name weight) Name weight) Embodiment 49 Perfluorinated 0.00001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.98499 acid polycarboxylic acid hydroxide water Embodiment 50 Perfluorinated 0.0001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9849 acid polycarboxylic acid hydroxide water Embodiment 8 Perfluorinated 0.001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 acid polycarboxylic acid hydroxide water Embodiment 51 Perfluorinated 0.01 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9750 acid polycarboxylic acid hydroxide water Embodiment 52 Perfluorinated 0.1 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.8850 acid polycarboxylic acid hydroxide water Comparative Perfluorinated 1 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 98.9850 Example 10 acid polycarboxylic acid hydroxide water

(29) TABLE-US-00011 TABLE 11 Surfactant Surfactant Alkali substance Distilled water Content Content Content Content (% by (% by (% by (% by Name weight) Name weight) Name weight) Name weight) Embodiment 53 Perfluorinated 0.00001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.98499 carboxylate polycarboxylic acid hydroxide water Embodiment 54 Perfluorinated 0.0001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9849 carboxylate polycarboxylic acid hydroxide water Embodiment 9 Perfluorinatrf 0.001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 carboxylate polycarboxylic acid hydroxide water Embodiment 55 Perfluorinated 0.01 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9750 carboxylate polycarboxylic acid hydroxide water Embodiment 56 Perfluorinated 0.1 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.8850 carboxylate polycarboxylic acid hydroxide water Comparative Perfluorinated 1 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 98.9850 Example 11 carboxylate polycarboxylic acid hydroxide water

(30) TABLE-US-00012 TABLE 12 Surfactant Surfactant Alkali substance Distilled water Content Content Content Content (% by (% by (% by (% by Name weight) Name weight) Name weight) Name weight) Embodiment 57 Perfluorianted 0.00001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.98499 sulfonate polycarboxylic acid hydroxide water Embodiment 58 Perfluorianted 0.0001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9849 sulfonate polycarboxylic acid hydroxide water Embodiment 10 Perfluorianted 0.001 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9840 sulfonate polycarboxylic acid hydroxide water Embodiment 59 Perfluorianted 0.01 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.9750 sulfonate polycarboxylic acid hydroxide water Embodiment 60 Perfluorianted 0.1 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 99.8850 sulfonate polycarboxylic acid hydroxide water Comparative Perfluorianted 1 Ammonium salt of 0.01 Tetrabutylammonium 0.005 Distilled 98.9850 Example 12 sulfonate polycarboxylic acid hydroxide water

Experimental Examples 1 to 60 and Comparative Experimental Examples 1 to 12

(31) Measurements of the pattern lifting defect level, the number-of-defects reduction ratio, the

transparency, the contact angle, and the surface tension were performed on the silicon wafers, the patterns on which were formed in Embodiments 1 to 60 and Comparative Examples 1 to 12. The measurements are described as Experimental Examples 1 to 60 and Comparative Experimental Examples 1 to 12. The results of the measurements are shown in Table 13.
(32) (1) Verification of Pattern Lifting Prevention
(33) After exposure energy and focus were split, among a total of 89 blocks, the number of blocks in which a pattern dis not collapse was measured using the critical dimension-scanning electron microscope (CD-SEM, manufactured by Hitachi, Ltd).
(34) (2) Number of Lifting Defects
(35) Counting of the number A of defects was performed on a photoresist pattern that was rinsed with each process liquid composition sample, using the surface defect observation equipment (manufactured by KLA-Tencor Corporation). A value of 100 was assigned to the number B of defects that resulted when the photoresist pattern was rinsed only with pure water. Then, the number A of defects was expressed as a ratio to the number B of defects, that is, as (A/B)×100.
(36) (3) Transparency
(37) Transparency of the manufactured process liquid composition was checked with the naked eye and was marked as a transparent or opaque process liquid composition.
(38) (4) Contact Angle and Surface Tension
(39) A surface tension and a contact angle of each of the process liquid compositions were measured using the contact angle measuring instrument [the DSA-100 Drop Shape Analyzer manufactured by KRÜSS GmbH] and the surface tension measuring instrument [the K100 Force Tensiometer manufactured by KRÜSS GmbH].
(40) TABLE-US-00013 TABLE 13 The number Number- of blocks of-defects with no reduction Contact Surface lifting ratio Trans- angle tension defects (%) parency (°) (mN/m) Experimental 80 25 transparent 49 22 Example 1 Experimental 78 30 transparent 55 25 Example 2 Experimental 78 35 transparent 52 22 Example 3 Experimental 77 35 transparent 57 23 Example 4 Experimental 77 40 transparent 56 23 Example 5 Experimental 75 55 transparent 57 26 Example 6 Experimental 72 62 transparent 58 30 Example 7 Experimental 73 60 transparent 52 29 Example 8 Experimental 76 55 transparent 54 27 Example 9 Experimental 75 48 transparent 53 25 Example 10 Experimental 77 30 transparent 50 22 Example 11 Experimental 78 38 transparent 55 28 Example 12 Experimental 78 42 transparent 53 24 Example 13 Experimental 77 44 transparent 58 27 Example 14 Experimental 76 58 transparent 56 26 Example 15 Experimental 76 64 transparent 58 30 Example 16 Experimental 71 74 transparent 59 33 Example 17 Experimental 72 59 transparent 53 32 Example 18 Experimental 74 61 transparent 54 29 Example 19 Experimental 74 52 transparent 54 28 Example 20 Experimental 64 44 transparent 60 39 Example 21 Experimental 70 38 transparent 55 29 Example 22 Experimental 77 28 transparent 42 19 Example 23 Experimental 76 30 transparent 36 15 Example 24 Experimental 58 50 transparent 59 40 Example 25 Experimental 65 36 transparent 57 33 Example 26 Experimental 75 32 transparent 52 22 Example 27 Experimental 74 33 transparent 49 19 Example 28 Experimental 60 55 transparent 59 37 Example 29 Experimental 68 42 transparent 56 28 Example 30 Experimental 75 38 transparent 49 19 Example 31 Experimental 73 39 transparent 45 16 Example 32 Experimental 61 52 transparent 60 36 Example 33 Experimental 67 44 transparent 58 30 Example 34 Experimental 74 37 transparent 54 21 Example 35 Experimental 73 38 transparent 50 17 Example 36 Experimental 60 60 transparent 59 37 Example 37 Experimental 65 49 transparent 57 31 Example 38 Experimental 73 42 transparent 54 20 Example 39 Experimental 73 44 transparent 52 17 Example 40 Experimental 59 77 transparent 60 38 Example 41 Experimental 67 65 transparent 58 32 Example 42 Experimental 71 58 transparent 55 23 Example 43 Experimental 70 60 transparent 53 20 Example 44 Experimental 52 80 transparent 60 40 Example 45 Experimental 60 69 transparent 60 34 Example 46 Experimental 68 64 transparent 56 27 Example 47 Experimental 66 65 transparent 55 24 Example 48 Experimental 61 82 transparent 57 38 Example 49 Experimental 69 73 transparent

55 35 Example 50 Experimental 72 62 transparent 50 24 Example 51 Experimental 71 65 transparent 48 20 Example 52 Experimental 63 68 transparent 58 39 Example 53 Experimental 70 62 transparent 56 33 Example 54 Experimental 74 55 transparent 51 23 Example 55 Experimental 73 56 transparent 50 22 Example 56 Experimental 61 57 transparent 60 38 Example 57 Experimental 68 50 transparent 55 29 Example 58 Experimental 72 49 transparent 50 20 Example 59 Experimental 71 50 transparent 48 18 Example 60 Comparative 46 100 transparent 89 70 Experimental Example 1 Comparative 40 95 transparent 58 67 Experimental Example 2 Comparative 58 150 transparent 35 14 Experimental Example 3 Comparative 54 172 transparent 50 19 Experimental Example 4 Comparative 52 184 transparent 44 16 Experimental Example 5 Comparative 51 186 opaque 47 16 Experimental Example 6 Comparative 50 180 opaque 49 16 Experimental Example 7 Comparative 51 210 opaque 53 20 Experimental Example 8 Comparative 50 235 opaque 54 22 Experimental Example 9 Comparative 50 170 opaque 46 38 Experimental Example 10 Comparative 52 168 opaque 49 21 Experimental Example 11 Comparative 51 174 opaque 47 18 Experimental Example 12

(41) From the comparison of Experimental examples 1 to 60 with Comparative Experimental examples 1 to 12, it could be seen that, when the number of blocks in which a pattern did not collapse was 50 or greater and the number-of-defects reduction ratio was 90% or less, a more improved result were obtained than in Comparative Experimental Example 1.

(42) It could be seen that the pattern lifting defect level was much more alleviated and the number of defects was much more reduced in Experimental Examples 1 to 60 than in Comparative Experimental Examples 1 to 12. The process liquid composition that was used in Experimental Examples 1 to 60 contained: 0.00001 to 0.1% by weight of a fluorine-based surfactant selected from among fluoroacryl carboxylate, fluoroalkyl ether, fluoroalkylene ether, fluoroalkyl sulfate, fluoroalkyl phosphate, fluoroacryl co-polymer, fluoro co-polymer, perfluorinated acid, perfluorinated carboxylate, and perfluorianted sulfonate; 0.0001 to 0.1% by weight of an anionic surfactant selected from among polycarboxylate salt, sulfonate salt, sulfate ester salt, and phosphoric acid ester salt; 0.0001 to 0.1% by weight of an alkali substance selected from among tetraethylammonium hydroxide, tetrapropylammonium hydroxide, and tetrabutylammonium hydroxide; and 99.7 to 99.99979% by weight of water.

(43) In addition, it could be seen that, desirably, effects of alleviating the pattern lifting defect level and reducing the number of defects was much more increased in the experimental examples 1 to 60 than in Comparative Experimental Examples 1 to 12. The process liquid composition that was used in Experimental Examples 1 to 60 contained: 0.0001 to 0.1% by weight of a fluorine-based surfactant selected from among fluoroacryl carboxylate, fluoroalkyl ether, fluoroalkylene ether, fluoroalkyl sulfate, fluoroalkyl phosphate, fluoroacryl co-polymer, fluoro co-polymer, perfluorinated acid, perfluorinated carboxylate, and perfluorianted sulfonate; 0.001 to 0.1% by weight of a hydrocarbon-based anionic surfactant selected from among polycarboxylate salt, sulfonate salt, sulfate ester salt, and phosphoric acid ester salt; 0.001 to 0.1% by weight of an alkali substance selected from among tetraethylammonium hydroxide, tetrapropylammonium hydroxide, and tetrabutylammonium hydroxide; and 99.7 to 99.9979% by weight of water.

(44) In addition, it could be seen that, more desirably, effects of alleviating the pattern lifting defect level and reducing the number of defects was much more increased in the experimental examples 1 to 60 than in Comparative Experimental Examples 1 to 12. The process liquid composition that was used in Experimental Examples 1 to 60 contained: 0.001 to 0.1% by weight of a fluorine-based surfactant selected from among fluoroacryl carboxylate, fluoroalkyl ether, fluoroalkylene ether, fluoroalkyl sulfate, fluoroalkyl phosphate, fluoroacryl co-polymer, fluoro co-polymer, perfluorinated acid, perfluorinated carboxylate, and perfluorianted sulfonate; 0.01 to 0.1% by weight of a hydrocarbon-based anionic surfactant selected from among polycarboxylate salt, sulfonate salt, sulfate ester salt, and phosphoric acid ester salt; 0.01 to 0.1% by weight of an alkali substance selected from among tetraethylammonium hydroxide, tetrapropylammonium hydroxide,

and tetrabutylammonium hydroxide; and 99.7 to 99.979% by weight of water.
(45) The result of measuring the collapse level of the photoresist pattern in Embodiment 1 for evaluation, was that the number of blocks in which the pattern did not collapse was 80.
(46) The result of measuring the collapse level of the photoresist pattern in Comparative Experimental Example 1 for evaluation, was that the number of blocks in which the pattern did not collapse was 46.
(47) The specific aspects of the present invention are described in detail above. It would be apparent to a person of ordinary skill in the art to which the present invention pertains that this specific description is only for the desired embodiments and do not impose any limitation on the scope of the present invention. Therefore, a substantial scope and a scope equivalent thereto must be defined by the following claims.

Claims

1. A process liquid composition for alleviating a lifting defect level of a photoresist pattern and for reducing the number of defects of the photoresist pattern, the composition containing a surfactant, and having a surface tension of 40 mN/m or less and a contact angle of 60° or smaller in the photoresist pattern having hydrophobicity represented by a contact angle of 70° or greater of water with respect to a photoresist surface in a photoresist pattern process, the composition comprising: 0.00001 to 0.1% by weight of a fluorine-based surfactant; 0.0001 to 0.1% by weight of a hydrocarbon-based anionic surfactant; 0.0001 to 0.1% by weight of an alkali substance that is selected from a group consisting of tetraethylammonium hydroxide, tetrapropylammonium hydroxide, tetrabutylammonium hydroxide, and mixtures thereof; and a remaining percentage by weight of water.
2. The process liquid composition of claim 1, comprising: 0.0001 to 0.1% by weight of the fluorine-based surfactant; 0.001 to 0.1% by weight of the hydrocarbon-based anionic surfactant; 0.001 to 0.1% by weight of the alkali substance; and 99.7 to 99.9979% by weight of the water.
3. The process liquid composition of claim 2, wherein the fluorine-based surfactant is selected from the group consisting of fluoroacryl carboxylate, fluoroalkyl ether, fluoroalkylene ether, fluoroalkyl sulfate, fluoroalkyl phosphate, fluoroacryl co-polymer, fluoro co-polymer, perfluorinated acid, perfluorinated carboxylate, perfluorianted sulfonate, and mixtures thereof.
4. The process liquid composition of claim 2, wherein the hydrocarbon-based anionic surfactant is selected from the group consisting of ammonium salt of polycarboxylic acid, sulfonate salt, sulfate ester salt, phosphoric acid ester salt, and mixtures thereof.
5. A method of forming a photoresist pattern, the method comprising: (a) dispensing photoresist on a semiconductor substrate and forming a photoresist film; (b) exposing the photoresist film to light, and developing the photoresist film to form a photoresist pattern; and (c) cleaning the photoresist pattern with the process liquid composition of claim 1.